Enhancing Video-LLM Reasoning via Agent-of-Thoughts Distillation

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Abstract

Video question answering demands multi-step reasoning and precise spatial-temporal grounding, capabilities that many Video Large Language Models (Video-LLMs) lack despite strong benchmark scores. Agent-based systems offer interpretable, tool-grounded reasoning but are slow and brittle at inference time; end-to-end Video-LLMs are efficient but opaque and error-prone on complex, compositional queries. We propose Agent-of-Thoughts Distillation (AoTD), a framework that bridges these two paradigms by distilling reliable, tool-grounded reasoning chains into a Video-LLM to jointly learn answers and rationales. AoTD builds an agent system that decomposes video queries into sub-tasks—temporal grounding, object detection, action recognition, and single-hop QA—and executes them with best-in-class specialist models, producing execution traces. These traces are converted into natural-language Chain-of-Thoughts (CoTs) and verified by an LLM for correctness and logical coherence. The filtered CoTs are then distilled into a Video-LLM via dual-prompt instruction tuning, jointly optimizing over answers and rationales. Experiments show that AoTD consistently improves LLaVA-NeXT-Video on both multiple-choice and open-ended VideoQA benchmarks (e.g., MVBench: 53.4% to 55.6%, ANetQA: 47.1% to 53.9%), while reducing inference latency from 47.93s to 10.58s and memory from 65.34GB to 18.21GB compared to the agent system. AoTD is model-agnostic and transfers effectively to Onevision, VideoLLaMA2, and Qwen2-VL.

1. Introduction

Video question answering (VideoQA) requires a model to perceive fine-grained spatial-temporal dynamics, perform multi-step compositional reasoning, and produce accurate, grounded answers. Recent Video Large Language Models such as LLaMA-VID [19], Video-LLaVA [20], VideoChat2 [17], VideoLLaMA2 [4], and LLaVA-NeXT-Video [47] have achieved impressive results by training large vision-language models end-to-end. However, these models often struggle on challenging compositional and temporal queries, generating answers in a single forward pass without any explicit step-by-step reasoning.

A complementary paradigm addresses complex visual queries through agent-based systems [7, 13, 14, 25, 34], where a controller decomposes the question into sub-tasks and invokes specialized vision tools such as object detectors, temporal grounders, and action recognizers. Such systems produce explicit execution traces that are interpretable and verifiable. However, they suffer from high inference latency due to repeated model invocations, large memory footprints from loading multiple specialist models, and brittleness to imperfect code generation.

We observe that these two paradigms are complementary: agent systems produce high-quality, step-wise reasoning traces but are slow; Video-LLMs are fast but lack explicit reasoning. This motivates a natural question: *can we distill the interpretable reasoning of an agent system into a single Video-LLM?*

We answer this affirmatively with **Agent-of-Thoughts Distillation (AoTD)**, as illustrated in Figure 1. AoTD operates in three main stages. First, an agent system decomposes each video query into sub-tasks and executes them using best-performing specialist models, collecting intermediate results and execution traces. Second, these program traces are converted into fluent natural-language CoTs via an LLM-based transfer step, then filtered by an LLM verifier to retain only correct and coherent rationales. Third, the verified CoTs are used to instruction-tune a Video-LLM using dual prompts—one requesting only the final answer and one requesting step-by-step reasoning—jointly optimizing both objectives. The full pipeline is depicted in Figure 2.

Our main contributions are:

- We propose **AoTD**, a distillation framework that imbues Video-LLMs with multi-step spatial-temporal reasoning by learning from agent-generated, LLM-verified CoTs.
- We construct an agent system that selects top-performing specialist models per sub-task via systematic evaluation, and converts execution traces into natural-language CoTs.
- We curate and filter a dataset of **32.3K high-quality CoTs** across compositional and temporal VideoQA

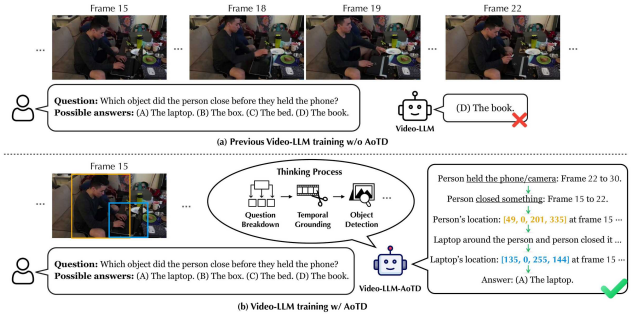

Figure 1. **Overview of AoTD.** Our method distills multi-step reasoning and spatial-temporal understanding into a single generative Video-LLM, achieving both the interpretability of agent systems and the efficiency of end-to-end models.

benchmarks for instruction tuning.

- We demonstrate consistent gains over strong baselines on MC-VQA and OE-VQA benchmarks, with emergent spatial-temporal grounding and markedly reduced inference cost versus the agent system.

2. Related Work

Video Large Language Models. Building on large vision-language models [1, 16, 22], a growing body of work extends them to video. Video-LLaVA [20] aligns video and language in a shared feature space. LLaMA-VID [19] represents each frame with two tokens for efficient long-video processing. VideoChat2 [17] introduces instruction tuning on large-scale video data and the MVBench suite. VideoLLaMA2 [4] adopts a spatiotemporal convolution connector for richer temporal encoding. LLaVA-NeXT-Video [47] extends the LLaVA-NeXT architecture to video by sampling frames uniformly. Despite their advances, these models generate answers in a single pass and lack explicit compositional reasoning and spatial-temporal grounding capabilities.

Agent-based Visual Question Answering. Agent systems tackle complex visual queries by decomposing them into sub-tasks and invoking specialized tools. ViperGPT [34] uses GPT-4 to generate Python programs that call vision APIs for image QA. VideoAgent [7] adapts this paradigm to video by iteratively querying relevant frames. AVIS [14] learns an adaptive visual information-seeking strategy. VPD [13] focuses on visual program distillation for images. MoreVQA [25] decomposes video queries modularly. DoraemonGPT [41] builds a comprehensive video agent with memory and planning. ProViQ [5] generates programs for video QA. While interpretable, these systems incur heavy inference costs from sequential model calls, which AoTD addresses by distilling their reasoning into a single efficient model.

Chain-of-Thought Reasoning for VLMs. Chain-of-Thought (CoT) prompting [36] has proven effective for multi-step reasoning in language models. VisCOT [33] applies visual CoT to vision-language models for grounded reasoning. CCOT [28] generates compositional CoTs for image understanding. Orca [27] distills step-by-step reasoning traces from GPT-4 into smaller models. mCoT [48] explores multimodal CoT for science QA. VideoThought [8] explores reasoning chains for video understanding. VSI-Bench [40] evaluates spatial reasoning in video. Unlike prior work, AoTD generates CoTs from agent execution traces for video, providing spatial-temporal grounding that is verified before distillation.

Knowledge Distillation for Vision-Language Models. Knowledge distillation transfers capabilities from large teacher models into compact students [**?**]. For language models, Orca [27] distills explanation traces from GPT-4 to improve smaller-model reasoning without requiring proprietary inference at test time. For VLMs, prior efforts focus on distilling image recognition or captioning capabilities. AoTD takes a distinct approach: the teacher is a heterogeneous multi-model agent system rather than a single larger network, and the distilled signal consists of grounded tool-execution traces that encode spatial-temporal structure unavailable from any single generative teacher model alone. This design is especially suited to VideoQA, where specialist perception models substantially outperform generalist VLMs on sub-tasks such as temporal grounding and object localization.

Compositional and Temporal VideoQA. Several benchmarks probe compositional and temporal reasoning. STAR [38] focuses on situated reasoning with four types. NExT-QA [39] tests causal and temporal question answering. AGQA [12] probes compositional spatial-temporal relations. ANetQA [45] evaluates diverse question types over ActivityNet videos. CLEVRER [42] tests spatial-temporal causal reasoning. EgoQA [11] covers egocentric video. VideoStar [49] further explores spatio-temporal reasoning. AoTD targets these challenging benchmarks by instilling explicit compositional reasoning into Video-LLMs.

3. Agent-of-Thoughts Distillation

Figure 2 provides a full overview of AoTD, comprising (1) agent-based CoT construction, (2) CoT transfer from traces to natural language, (3) LLM-based filtering, and (4) dual-prompt distillation.

3.1. Agent-based CoT Construction

Sub-task Decomposition. Given a video V and a query Q (with possible answers $\mathcal{A}$ for MC settings), the agent system

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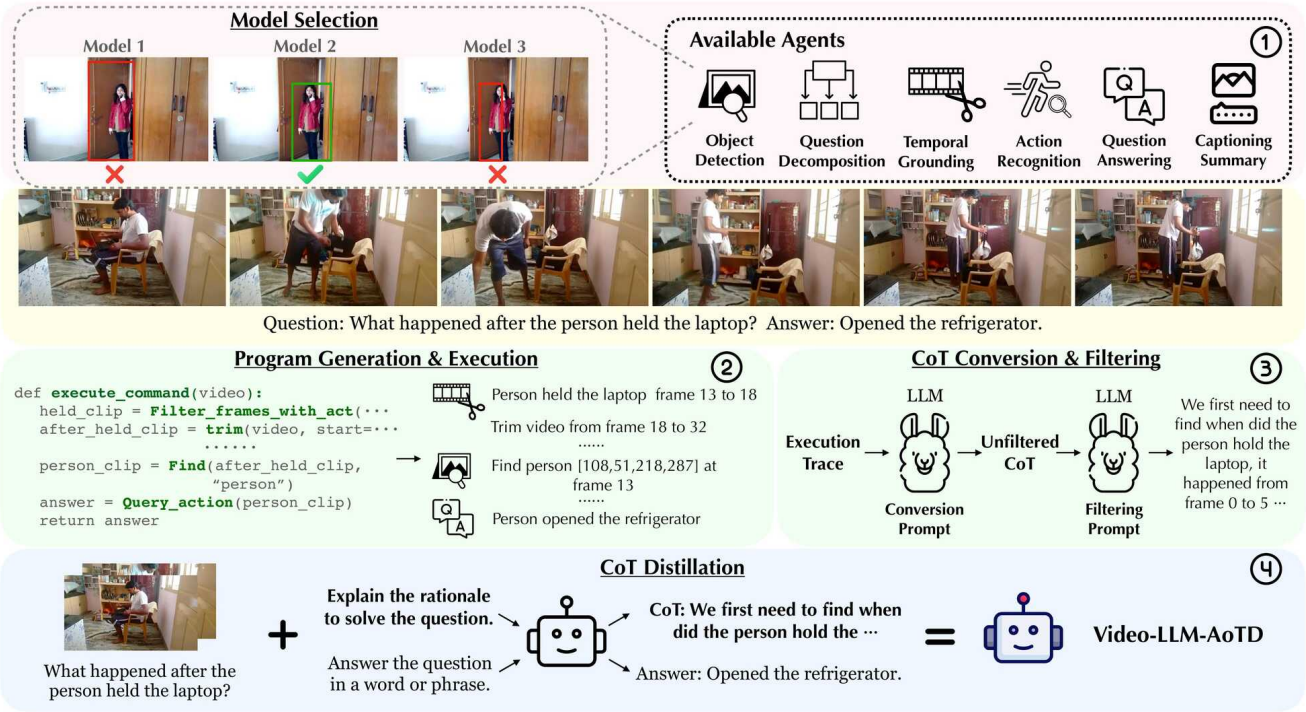

Figure 2. **AoTD pipeline.** Given a video and query, the agent system decomposes the query into sub-tasks, executes them with specialist models, and collects execution traces. An LLM converts these traces into natural-language CoTs, which are then verified and used to instruction-tune a Video-LLM via dual-prompt distillation.

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uses a code-generation LLM to produce a Python program that decomposes Q into a sequence of specialized sub-tasks. We define five sub-task types (Table 1):

1. **Question decomposition:** Parse the complex query into simpler sub-questions to be answered sequentially.
2. **Object detection:** Localize and identify objects in relevant video frames.
3. **Temporal grounding:** Identify the temporal segment relevant to a sub-question.
4. **Action recognition:** Classify the action or activity in a given clip.
5. **Question answering:** Answer a simple single-hop sub-question given visual or temporal context.

Specialist Model Selection. For each sub-task type, we evaluate multiple candidate models on the STAR training set and select the best performer (Table 1):

- *Question decomposition:* DeepSeek-Coder-Instruct (6.7B) [6] achieves 85.7% accuracy, outperforming CodeQwen1.5-Chat (52.7%) and GPT-3.5-Turbo (73.1%).
- *Object detection:* OWL-ViT v2 [26] achieves 63.0 IoU, outperforming OWL-ViT v1 (47.3) and GLIP [18] (58.9).
- *Temporal grounding:* UniVTG [21] achieves 24.7/35.3 IoU/Recall, outperforming LITA-13B [15] (11.7/20.2) and TimeChat-7B [32] (13.9/23.1).
- *Action recognition:* LLaVA-NeXT-Video-DPO (7B) [47] achieves 18.2% Top-1 accuracy, outperforming InternVideo2 [35] (7.6%) and Open-VCLIP [37] (8.9%).
- *Question answering:* LLaVA-NeXT-Video-DPO (7B) [47] achieves 53.4% accuracy, outperforming LLaMA-VID [19] (43.5%) and SeViLA [43] (46.5%).

Execution Trace Collection. The agent executes the generated program step-by-step, invoking selected specialist models and recording intermediate results. As shown in Figure 3, each execution produces a trace comprising (i) the generated program, (ii) intermediate outputs at each step (e.g., detected objects, temporal segments, sub-answers), and (iii) the final predicted answer. Only traces where the predicted answer matches the ground-truth label are retained.

3.2. CoT Transfer

Raw execution traces consist of program code and structured intermediate outputs, unsuitable for direct instruction tuning. We convert them into fluent natural-language rationales using LLaMA-3.1-8B-Instruct with a dedicated transfer prompt. This prompt instructs the LLM to narrate the reasoning steps coherently, describing what each sub-task

Sub-task	Model name	Metric	Number (%)
question decomposition	CodeQwen1.5-Chat (7B) [2]	Acc	52.7
	GPT-3.5-Turbo [29]		73.1
	DeepSeek-Coder-Instruct (6.7B) [6]		85.7
object detection	OWL-ViT v1 [24]	IoU	47.3
	GLIP [18]		58.9
	OWL-ViT v2 [26]		63.0
temporal grounding	LITA (13B) [15]	IoU / Recall	11.7 / 20.2
	TimeChat (7B) [32]		13.9 / 23.1
	UniVTG [21]		24.7 / 35.3
action recognition	InternVideo2 (1B) [35]	Top1-Acc	7.6
	Open-VCLIP [37]		8.9
	LLaVA-NeXT-Video-DPO (7B) [47]		18.2
question answering	LLaMA-VID (7B) [19]	Acc	43.5
	SeViLA [43]		46.5
	LLaVA-NeXT-Video-DPO (7B) [47]		53.4

Table 1. Sub-tasks definition and evaluation results. We choose 3 model candidates for each sub-task and evaluate them in STAR training set with the corresponding metrics. Models with best performance are placed at the bottom of each column.

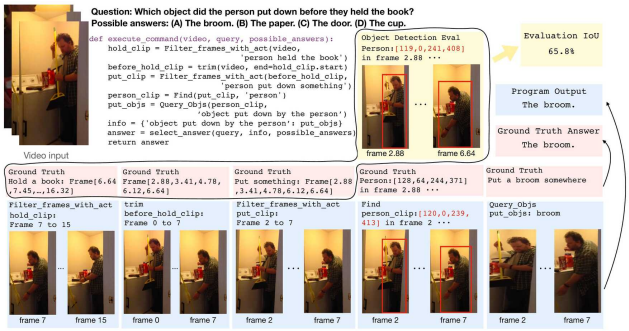

Figure 3. **Execution process.** An example of step-by-step program execution within the agent system, showing sub-task decomposition, specialist tool invocation, and intermediate result collection.

found and how intermediate results led to the final answer. The output is a natural-language CoT such as: "*First, I identify the temporal segment in which the action occurs... Then, I detect the objects present... Based on these observations, the answer is...*"

3.3. CoT Filtering

Not all transferred CoTs are logically coherent or faithfully reflect the video content. We apply LLM-based verification using LLaMA-3.1-8B-Instruct with a filtering prompt: given the question and candidate rationale, the LLM outputs `True` if the rationale is logically consistent and correctly leads to the given answer, or `False` otherwise. Only CoTs passing this filter are retained for training. Table 6 shows that filtering significantly improves performance (e.g., MVBench: 53.7% to 55.6%, STAR: 73.3% to 74.3%).

The filtering prompt supplies the LLM with the original query, the candidate CoT, and the ground-truth answer, asking it to verify three criteria: (i) *factual consistency*—each sub-task output is plausible given the query; (ii) *logical coherence*—intermediate results are correctly chained toward a conclusion; and (iii) *answer alignment*—the stated conclusion matches the ground-truth label. A CoT is retained only when all three criteria are met, ensuring that distilled rationales encode reliable spatial-temporal evidence rather than spurious or hallucinated reasoning steps.

3.4. Dual-Prompt Distillation

The filtered CoTs are used to instruction-tune a Video-LLM via dual-prompt training, inspired by knowledge distillation [27]. Each training sample appears in two variants:

- **Answer prompt:** The question and options with the suffix "*Answer with the option's letter from the given choices directly.*" Target: the ground-truth answer letter.
- **Rationale prompt:** The same question and options with the suffix "*Explain the rationale to answer the question.*" Target: the filtered CoT.

Both variants are included in each training batch. The model is fine-tuned with standard cross-entropy loss over target tokens, jointly optimizing answer prediction and rationale generation. This dual objective encourages the model to internalize step-wise spatial-temporal reasoning while retaining concise-answer capability. We fine-tune LLaVA-NeXT-Video-7B [47] with a maximum sequence length of 8192 tokens and a batch size of 4.

3.5. Training Data

We construct CoTs for six VideoQA datasets covering complementary reasoning skills (Table 2). For MC benchmarks: STAR [38] (compositional, 45.7K labels, 11.2K CoTs), NExT-QA [39] (temporal and causal, 34.1K labels, 12.1K CoTs), CLEVRER [42] (spatial-temporal, 21.0K labels). For OE benchmarks: AGQA [12] (compositional, 25.0K labels, 5.4K CoTs), ANetQA [45] (compositional, 25.0K labels, 3.6K CoTs), and EgoQA [11] (egocentric, 7.8K labels). In total we process 158.6K VideoQA pairs and generate **32.3K high-quality CoTs** after filtering. The per-dataset retention rates vary considerably: STAR retains 24.5% of executed traces, NExT-QA 35.5%, and ANetQA 14.4%, reflecting that compositional multi-hop queries tend to produce more logically coherent rationales than temporally complex open-ended questions where the agent's sub-task outputs are noisier.

4. Experiments

4.1. Experimental Setup

Base Model and Baselines. Our primary model is LLaVA-NeXT-Video-7B [47], fine-tuned with AoTD to

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Dataset	Description	# Labels	# CoTs
AGQA	Compositional	25.0K	5.4K
ANetQA	Compositional	25.0K	3.6K
STAR	Compositional	45.7K	11.2K
NExT-QA	Temporal & Causal	34.1K	12.1K
CLEVRER	Spatial & Temporal	21.0K	-
EgoQA	Ego-centric	7.8K	-
Total		**158.6K**	**32.3K**

Table 2. Dataset statistics. The column "# Labels" indicates the number of VideoQA pairs, which include the video, query, possible answers (multiple-choice), and the correct answer. "# CoTs" refers to the number of CoTs generated using our agent-based system for each dataset.

obtain **LLaVA-NeXT-Video-AoTD** (LNV-AoTD). The main baseline is **LLaVA-NeXT-Video-Instruct** (LNV-Instruct), instruction-tuned on the same datasets but without CoT rationales. We also compare against VideoLLaMA [46], Video-ChatGPT [23], LLaMA-VID [19], Video-LLaVA [20], VideoChat2 [17], VideoLLaMA2 [4], and proprietary models Gemini 1.5 Pro [10] and GPT-4V [30].

Evaluation Benchmarks. We evaluate on the datasets in Table 3. For MC-VQA: MVBench [17], VideoMME [9], STAR [38], NExT-QA [39], Perception-Test [31], and VSI-Bench [40]. For OE-VQA: ANetQA [45], AGQA [12], ActivityNet-QA [44], and Video-ChatGPT [23].

Implementation Details. We uniformly sample 32 frames per video. We use the AdamW optimizer with a peak learning rate of 2e-5, weight decay of 0.05, and linear warmup over 3% of total steps followed by cosine decay. Training runs for 1 epoch on 8 NVIDIA A100 (80GB) GPUs with gradient checkpointing enabled to handle the extended sequence lengths introduced by CoT rationales. Both answer and rationale prompt variants are included in every training batch.

4.2. Main Results on MC-VQA

Table 4 reports results on six MC-VQA benchmarks. LNV-AoTD achieves consistent gains over LNV-Instruct across all benchmarks: MVBench (+2.2%), VideoMME (+1.8%), STAR (+2.1%), NExT-QA (+0.5%), Perception-Test (+0.3%), and VSIBench (+2.1%). Compared to other 7B open-source models, LNV-AoTD outperforms VideoLLaMA2 on MVBench (53.4 vs. 55.6) and VideoChat2 (51.1 vs. 55.6) while using the same backbone architecture.

4.3. Main Results on OE-VQA

Table 5 shows results on OE-VQA benchmarks. LNV-AoTD achieves substantial gains over LNV-Instruct on

Dataset	Size		Type	Train	Eval
	train	eval			
MC-VQA					
STAR [38]	45.7K	7.1K	Compositional	✓	✓
NExT-QA [39]	34.1K	5.0K	Temporal & Causal	✓	✓
CLEVRER [42]	21.0K	-	Spatial-temporal	✓	✗
Perception-Test [31]	-	11.5K	General	✗	✓
MVBench [17]	-	4.0K	General	✗	✓
VideoMME [9]	-	2.7K	General	✗	✓
VSIBench [40]	-	5.0K	spatial-temporal	✗	✓
OE-VQA					
AGQA [12]	25.0K	2.0K	Compositional	✓	✓
ANetQA [45]	25.0K	2.0K	Compositional	✓	✓
EgoQA [11]	7.8K	-	Ego-centric	✓	✗
Activitynet-QA [44]	-	8.0K	General	✗	✓
Video-ChatGPT [23]	-	3.0K	General	✗	✓

Table 3. Training and evaluation datasets statics.

ANetQA (47.1/3.1 to 53.9/3.4 Acc/Score) and ActivityNet-QA (50.0/3.3 to 53.2/3.4). On AGQA, accuracy improves from 59.3 to 60.9. Video-ChatGPT metrics similarly improve: Correctness (2.96 to 3.11), Detail (2.81 to 3.00), and Context (3.35 to 3.60), demonstrating that distilled rationales enrich the model's descriptive capabilities.

4.4. Ablation Studies

Effect of CoT Filtering. Table 6 (upper section) compares LNV-AoTD with and without filtering. Removing filtering degrades performance on all benchmarks: MVBench drops from 55.6% to 53.7%, STAR from 74.3% to 73.3%, and AGQA from 60.9/3.6 to 59.5/3.5 (Acc/Score). This confirms that noisy or incoherent CoTs are harmful, and LLM-based verification is essential.

Effect of Dual-Prompt Training. We ablate the two training objectives in the dual-prompt strategy by training with only the answer prompt (Answer-Only) or only the rationale prompt (Rationale-Only) and comparing to the full dual-prompt AoTD. Answer-Only fine-tuning, which omits rationale supervision, yields MVBench 54.1% and STAR 72.6%—below LNV-AoTD (55.6% / 74.3%), confirming that CoT supervision contributes beyond data augmentation. Rationale-Only training, which never explicitly trains the concise-answer path, achieves MVBench 54.8% and STAR 73.1%, slightly worse on OE-VQA due to format mismatch. The full dual-prompt objective consistently outperforms both ablations, demonstrating that jointly optimizing answer accuracy and rationale generation produces a synergistic benefit: rationale training regularizes the answer head toward structured reasoning, while answer training prevents the model from producing verbose outputs at the expense of correctness.

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Model	MVBench (Acc.)	VideoMME (Acc.)	STAR (Acc.)	NExT-QA (Acc.)	Perception-Test (Acc.)	VSIBench (Acc.)
Proprietary Models						
Gemini 1.5 Pro [10]	-	75.0	-	-	-	45.4
GPT4-V [30]	43.7	59.9	-	-	-	-
Open-source Models						
LLaMA-VID (7B) [19]	41.9	25.9	-	-	44.6	-
Video-LLaVA (7B) [20]	41.0	39.9	-	-	44.3	-
VideoChat2 (7B) [17]	51.1	33.7	59.0*	68.6*	47.3	-
VideoLLaMA2 (7B) [4]	53.4	45.1	58.5*	62.3*	49.6	-
LLaVA-NeXT-Video (7B) [47]	46.5*	41.0*	52.4*	61.6*	47.5*	19.7*
LLaVA-NeXT-Video-Instruct (7B)	53.4	43.2	72.2	77.1	50.3	26.7
LLaVA-NeXT-Video-AoTD (7B)	55.6	45.0	74.3	77.6	50.6	28.8

Table 4. Comparison with Video-LLMs on MC-VQA benchmarks. LLaVA-NeXT-Video-AoTD outperforms all other baselines the and the version without CoT distillation. * means results reproduced by ourseleves. Results without signs are retrieved from [4] and [40].

Model	ANetQA (Acc./Score)	AGQA (Acc./Score)	Video-ChatGPT (Score)					ActivityNet (Acc./Score)
			Corr.	Deta.	Cont.	Temp.	Cons.	
Proprietary Models								
Gemini 1.5 Pro [10]	-	-	-	-	-	-	-	56.7/-
GPT4-V [30]	-	-	4.09	3.88	4.37	3.94	4.02	59.5/-
Open-Source Models								
VideoLLaMA (7B) [46]	-	-	1.96	2.18	2.16	1.82	1.79	12.4/1.1
Video-ChatGPT (7B) [23]	-	-	2.50	2.57	2.69	2.16	2.20	35.2/2.7
LLaMA-VID (7B) [19]	-	-	2.96	3.00	3.53	2.46	2.51	47.4/3.3
Video-LLaVA (7B) [20]	-	-	2.87	2.94	3.44	2.45	2.51	45.3/3.3
VideoChat2 (7B) [17]	-	-	3.02	2.88	3.51	2.66	2.81	49.1/3.3
VideoLLaMA2 (7B) [4]	-	-	3.09	3.09	3.68	2.63	3.25	49.9/3.3
LLaVA-NeXT-Video (7B) [47]	46.4/3.3*	27.4/2.2*	3.26*	3.22*	3.77*	2.47*	2.99*	54.3/3.2*
LLaVA-NeXT-Video-Instruct (7B)	47.1/3.1	59.3/3.4	2.96	2.81	3.35	2.42	2.82	50.0/3.3
LLaVA-NeXT-Video-AoTD (7B)	53.9/3.4	60.9/3.6	3.11	3.00	3.60	2.41	2.91	53.2/3.4

Table 5. Comparison with Video-LLMs on OE-VQA benchmarks. LLaVA-NeXT-Video-AoTD improves performance in all open-ended benchmarks compared with the Instruct version. * means results reproduced by ourseleves. Results without signs are retrieved from [4].

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Effect of CoT Data Scale. Training LNV-AoTD with 25%, 50%, and 100% of the 32.3K CoTs yields MVBench accuracies of 54.2%, 55.1%, and 55.6%, respectively. This monotonic improvement indicates that performance has not yet saturated, and collecting additional agent-verified CoTs remains a promising avenue for further gains.

Transferability to Other Video-LLMs. Table 6 (lower section) shows that our CoT dataset transfers to other Video-LLMs. Applying AoTD CoTs to Onevision [3] yields MVBench 59.2 to 60.5 and STAR 75.8 to 76.6. VideoLLaMA2 [4] similarly benefits (MVBench 54.9 to 56.0, STAR 69.2 to 71.1). Qwen2-VL improves on all three metrics. These results confirm that AoTD is model-agnostic.

Model	Filtering	MVBench (Acc.)	STAR (Acc.)	AGQA (Acc. / Score)
LNV-AoTD	✗	53.7	73.3	59.5/3.5
LNV-AoTD	✓	55.6	74.3	60.9/3.6
Onevision-Instruct	-	59.2	75.8	65.6/3.7
Onevision-AoTD	✓	60.5	76.6	65.7/3.7
VideoLLaMA2-Instruct	-	54.9	69.2	56.0/3.5
VideoLLaMA2-AoTD	✓	56.0	71.1	57.2/3.5
Qwen2-VL-Instruct	-	65.6	71.4	59.8/3.6
Qwen2-VL-AoTD	✓	66.5	73.1	61.2/3.7

Table 6. Ablation results of CoT filtering and transferability.

4.5. Efficiency Analysis

Table 7 compares the agent-based system and LNV-AoTD at inference time. The agent system requires 47.93s per sample (sequential model invocations), 65.34GB memory (multiple specialist models loaded simultaneously), and 30.98 TFLOPs. LNV-AoTD reduces latency to 10.58s (4.5× speedup), memory to 18.21GB (3.6× reduction), and computation to 13.53 TFLOPs (2.3× reduction). AoTD retains the reasoning quality of the agent system while achieving single-model efficiency. This efficiency gain is espe-

cially important for latency-sensitive applications such as video surveillance, autonomous driving assistance, or interactive video search, where the agent system's 48s-per-query cost would be entirely impractical for deployment.

Model	Time (s) ↓	Memory (GB) ↓	TFLOPs ↓
Agent-based system	47.93	65.34	30.98
LNV-AoTD	10.58	18.21	13.53

Table 7. Latency and computation comparison.

4.6. Spatial-Temporal Grounding Analysis

Table 8 evaluates grounding capabilities. LNV-Instruct cannot produce temporal or spatial predictions (indicated by ✗), as it lacks grounding supervision. In contrast, LNV-AoTD achieves temporal grounding IoU of 21.7% and Recall of 34.0%, comparable to the specialist UniVTG [21] (22.8 IoU / 31.0 Recall). For spatial grounding, LNV-AoTD achieves 45.2% IoU, below the specialist OWL-ViT v2 [26] (64.7%) but notable as an emergent capability from CoT distillation without explicit grounding supervision.

This emergent grounding behavior arises because rationale targets describe temporal and spatial observations in natural language, and the model learns to associate these descriptions with its internal visual representations during training. The grounding capacity emerges as a by-product of learning structured rationales rather than from dedicated localization supervision, suggesting that language-grounded CoTs implicitly encode perceptual structure that generalizes beyond QA to localization tasks.

4.7. Qualitative Analysis

Figure 4 presents qualitative examples of rationales generated by LNV-AoTD. The model correctly identifies relevant temporal segments, grounds objects in video frames, and chains these observations into coherent reasoning paths leading to correct answers. LNV-Instruct, without rationale training, directly produces answers and often fails on queries requiring multi-step spatial-temporal understanding. Importantly, errors made by LNV-AoTD tend to occur at specific reasoning steps—such as misidentifying a temporal boundary or an occluded object—rather than from a complete failure of structured reasoning. This makes failure modes transparent and diagnosable, a practical advantage over opaque single-pass predictions. The distilled rationales thus serve a dual purpose: improving accuracy and providing interpretable justifications that could support human oversight in deployment.

Error Analysis. We further categorize failure modes of LNV-AoTD to characterize its remaining limitations. Errors arise primarily in two scenarios: (i) queries requiring fine-grained counting or precise numerical reasoning, where the object-detection specialist produces imprecise localizations that propagate into the CoT and mislead subsequent reasoning steps; and (ii) questions about rare or domain-specific actions under-represented in training data, where the action-recognition specialist outputs low-confidence predictions that the CoT transfer step struggles to articulate coherently. Notably, LNV-AoTD tends to hedge rather than hallucinate confidently in these cases, suggesting that rationale supervision instills calibrated uncertainty alongside accuracy gains—a desirable property for safety-critical video understanding applications.

Model	Temporal Grounding IoU (%)	Temporal Grounding Recall (%)	Spatial Grounding IoU (%)
UniVTG	22.8	31.0	-
OWL-ViT v2	-	-	64.7
LNV-Instruct	✗	✗	✗
LNV-AoTD	21.7	34.0	45.2

Table 8. Temporal and spatial abilities evaluation results.

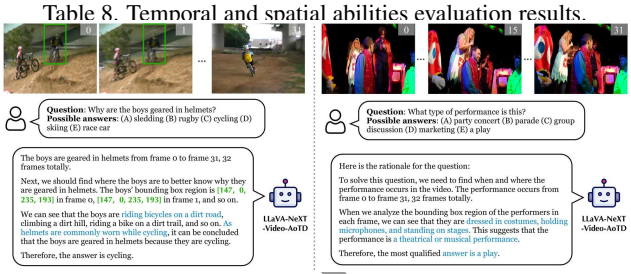

Figure 4. **Visualization of rationales.** Examples of step-by-step rationales generated by LNV-AoTD, demonstrating grounded temporal and spatial reasoning on VideoQA queries.

5. Conclusion

We presented **Agent-of-Thoughts Distillation (AoTD)**, a framework for enhancing Video-LLMs with multi-step spatial-temporal reasoning by distilling agent-generated Chain-of-Thoughts. AoTD constructs an agent system that decomposes video queries, invokes best-in-class specialist models, converts execution traces into natural-language rationales, verifies them with an LLM, and instruction-tunes a Video-LLM with dual answer-and-rationale prompts. Extensive experiments demonstrate consistent improvements over strong baselines on both MC-VQA and OE-VQA benchmarks, emergent spatial-temporal grounding capabilities, and dramatically reduced inference cost compared to agent systems. AoTD is model-agnostic and transfers effectively to Onevision, VideoLLaMA2, and Qwen2-VL, demonstrating the broad applicability of tool-grounded CoT distillation for video understanding. Current limitations include reliance on a capable code-generation LLM for subtask decomposition and higher training complexity relative to standard instruction tuning. Furthermore, the CoT filtering step discards approximately 80% of generated rationales overall, indicating that better trace-to-language con-

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version methods could substantially expand the training corpus and unlock additional performance gains without requiring new video annotations. Future work will explore extending AoTD to longer videos and egocentric scenarios, incorporating retrieval-augmented sub-task execution, and designing self-improvement loops in which the Video-LLM iteratively refines its own CoT quality without requiring the full agent system at each iteration.

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